

Course Code- 119406 Artificial Intelligence**3 0 0 3****Unit-1.0: Introduction:****7 hrs**

Overview, Turing test, Intelligent agents.

Unit-2.0: Problem Solving:**7 hrs**

Solving Problems by Searching: Uninformed search - Depth First Search, Breadth First Search, DFID, Heuristic search - Generate and Test, Best First Search, Beam Search, Hill Climbing, A*,

Unit-3.0:**7 hrs**

Problem reduction search – AND/OR Graphs, AO*, Constraint satisfaction, Means-ends analysis, Stochastic search methods - Simulated Annealing, Particle Swarm Optimization, Game Playing - Minimax algorithm, Alpha-beta pruning.

Unit-4.0: Knowledge and Reasoning:**7 hrs**

Building a knowledge base - Propositional logic, first order logic, Inference in first order logic, Resolution – refutation proofs, Theorem Proving in First Order Logic; Planning, partial order planning, Uncertain Knowledge and Reasoning, Probabilities, Bayesian Networks.

Unit-5.0: Learning:**7 hrs**

Overview of different forms of learning: unsupervised, supervised, semi-supervised, K-means clustering algorithm, Decision Trees, Neural Networks, Deep Learning.

Unit-6.0: Advanced topics:**7 hrs**

Introduction to Computer Vision, Natural Language Processing, Expert Systems, Robotics, Genetic Algorithm,

Text/ Reference:-

1. S. Russell and P. Norvig, "Artificial Intelligence: A Modern Approach," Prentice Hall
2. E. Rich, K. Knight and S. B. Nair, "Artificial Intelligence," TMH.
3. C. Bishop, "Pattern Recognition and Machine Learning," Springer
4. D. W. Patterson, "Introduction to artificial intelligence and expert systems," Prentice Hall
5. A. C. Staugaard, Jr., "Robotics and AI: An Introduction to Applied Machine Intelligence," Prentice Hall
6. I. Bratko, "Prolog Programming for Artificial Intelligence," Addison-Wesley
7. S. O. Haykin, "Neural Networks and Learning Machines," Prentice Hall
8. D. Jurafsky and J. H. Martin, "Speech and Language Processing," Prentice Hall.